

WASP-43b

R-1011

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_160501 · created 2026-07-30T02:58:54+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457509.67671 +/- 0.00099 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.146 +/- 0.019
Transit depth (Rp/Rs)^2	2.12 +/- 0.54 [%]
Orbital Inclination (inc)	83.43 +/- 1.64
Airmass coefficient 1 (a1)	0.957 +/- 0.014
Airmass coefficient 2 (a2)	0.0232 +/- 0.0059
Scatter in the residuals of the lightcurve fit is	1.48 %
Best Comparison Star	#2 - [106, 81]
Optimal Aperture	3.5
Optimal Annulus	10.55
Transit Duration (day)	0.0533 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.146 $\pm$ 0.019 vs 0.1594 (deviation 0.71 σ)
Duration fit vs published	0.0533 d vs 0.0512 d (deviation 0.48 σ)
Mid-time O-C	-2.0 min
Depth SNR	7.7
Residual scatter	1.48 %

Light curve

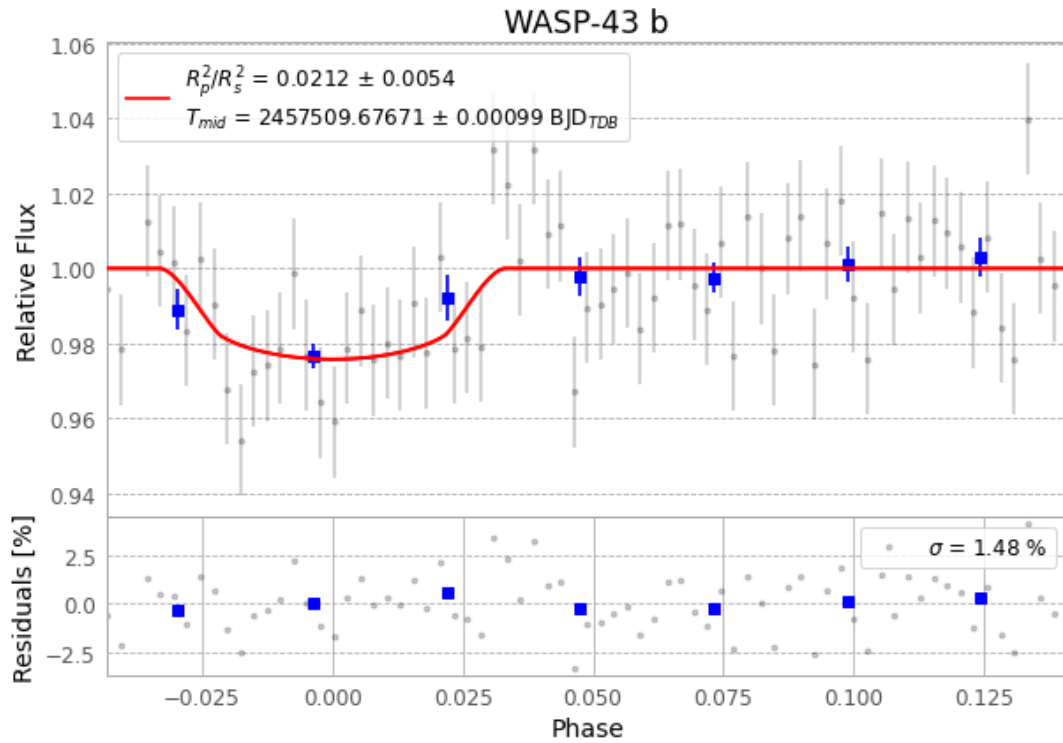


Figure 1 — FinalLightCurve_WASP-43 b_2016-04-30.png (SHA-256 c056ce4ecd1fe42a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [192, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b0060e74ed88d71d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

73 FITS frames (48 MB), WASP-43160501031812.FITS → WASP-43160501065412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-160501.log (53fa2d361669...), FinalLightCurve_WASP-43 b_2016-04-30.png (c056ce4ecd1f...), FinalLightCurve_WASP-43 b_2016-04-30.csv (169dc15c1e9b...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 02:58Z.